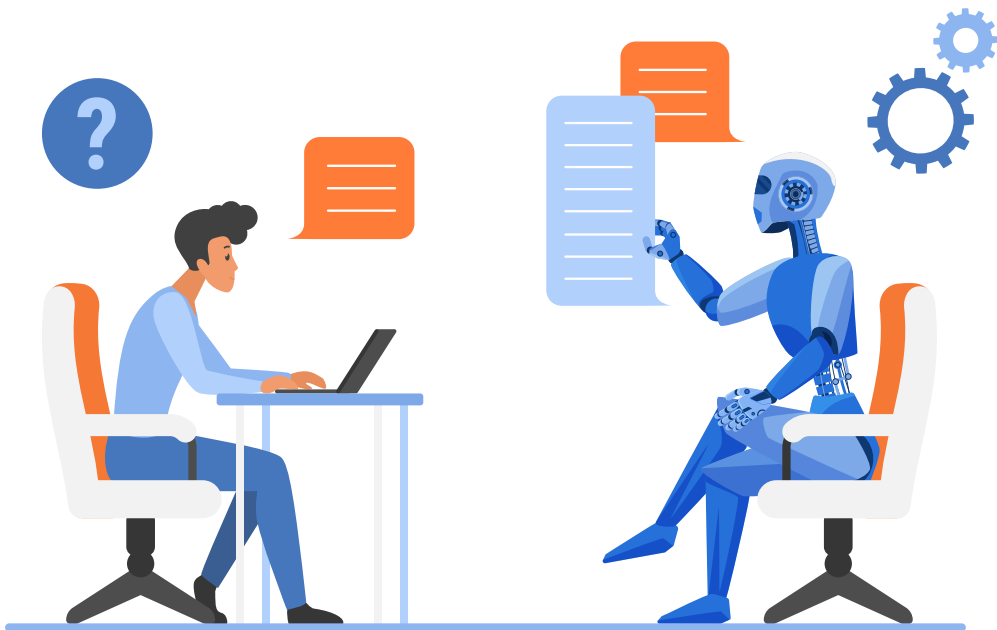




Interview Questions for AI Engineering



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A. Entry Level

1

"You're given a public open-source model like BERT and asked to fine-tune it for sentiment classification on customer reviews in a new domain. Walk me through how you'd adapt the dataset, choose hyper-parameters, monitor overfitting, and validate the model before deployment."

2

Explain how you would build a minimal inference endpoint for a deep-learning model (say image classification) in a cloud setting. What are the main bottlenecks (latency, throughput, cost) and how would you mitigate them?

3

You observe that after deploying your model you start getting a performance drop in real-world usage. How do you detect whether it's data drift or concept drift, and what concrete steps would you take to remediate it?

4

In an A/B test for a new recommendation model you created, the online metric improved but the offline test set performance dropped relative to the baseline. How do you reconcile this difference? What might be going on?

5

Describe how you would ensure interpretability of a model built with LightGBM when deployed in a regulated industry (e.g., finance). What tools or techniques would you use?



A. Entry Level

6

You need to deploy a model such that it can be rolled back safely, versioned, and monitored. Sketch the architecture of your model-serving stack including feature store, model registry, logging/metrics

7

Given an imbalanced dataset (say 1% positive class, 99% negative) where you train a binary classifier, what evaluation metrics would you pick and why? Also, what strategies would you use to improve the model?

8

A business partner asks: 'Why did the model make this prediction?', and they are not technical. How do you explain the reasoning in non-technical language, and how do you surface evidence from the model to support your explanation?

9

You are tasked with selecting between a pre-trained open-source model versus training from scratch on your data for a specific domain. What criteria and trade-offs guide your decision (compute cost, data size, domain shift, licensing)?

10

What's your strategy for hyper-parameter tuning of a neural network (e.g., learning rate schedules, batch size, regularization)? What trade-offs do you consider between search time, model performance, and reproducibility?



B. Intermediate Level

11

Design a system that uses retrieval-augmented generation (RAG) using an open-source embedding model (e.g., OpenAI Embeddings or similar) and a vector-database for customer support. Walk through data ingestion, embeddings, retrieval, LLM prompt chain, and how you evaluate correctness/hallucination risk in production.

12

You have an LLM-based agent that has access to multiple internal APIs (inventory lookup, order status, product catalog). How do you architect memory (short-term vs long-term) for the agent, how do you control invocation of APIs, and how do you handle failure/fallback when the API fails or returns unexpected data?

13

Describe how you would design a continuous-training pipeline for an ML model that serves live traffic, including how you'd choose when to retrain, how you'd validate the new model, how you'd roll it out (canary vs full), and how you'd monitor for degradation post-release.

14

Compare and contrast model compression techniques (quantization, pruning, distillation) for deploying a large transformer model on edge devices. Provide example trade-offs in latency, accuracy, memory and how you'd measure success.

15

You observe that your model produces biased outputs (e.g., gender or geography-based bias). Walk me through how you would detect, quantify, and mitigate this bias, and how you'd implement monitoring to guard against future bias drift.



B. Intermediate Level

16

Open-source model licensing and use-case constraints are increasingly important. Suppose you're evaluating an open-source LLM (e.g., under Apache 2.0 vs GPL) for integration into your product. What legal/operational/engineering risks do you assess, and how does this influence your model selection or architecture?

17

When designing the evaluation framework for a generative model (say summarisation or code generation), how do you define metrics (automated + human), set up red-teaming/hallucination tests, and build feedback loops to improve the model over time?

18

You need to embed an ML model in a micro-services architecture that serves high QPS (queries per second) with tight latency SLA. What infrastructure decisions (caching, batching, model sharding, autoscaling) do you make? How do you monitor and mitigate overload?

19

Consider a scenario where you must deploy a 'tool-invoking' agent (LLM calls external tools) at scale for internal enterprise use. How would you instrument for traceability, observability, for debugging tool-invocation chains, and what metrics would you track?

20

You are leading a team that contributes to an open-source ML framework. How do you decide which features to build, how do you incorporate community feedback, ensure quality, compatibility, and how do you measure the health of the project (bugs, pull-requests, adoption)?



C. Advanced Level

21

Design a large-scale evaluation infrastructure for benchmarking open-source LLMs on multiple tasks (e.g., reasoning, code, multilingual). How do you select tasks and datasets to avoid overfitting, how do you manage compute cost, how do you ensure reproducibility, and how do you report results fairly (including model size, latency, training cost)?

22

You're tasked with building an agentic system where an LLM uses a set of tool-modules (APIs), has a memory component and needs to reason over time. Describe the architecture: how do you structure the loop (observe > decide > act), how do you govern the memory lifetime, how do you ensure safety/fail-safe, how do you monitor for drift or goal-corruption of the agent over time?

23

Suppose you have a proprietary model and an open-source model that performs similarly but the proprietary model has higher cost and locked-down usage. How would you set up an experimental design (including statistical tests, power, decoy control, production hold-out) to decide which model to push to production, and how would you guard the decision against business risk (cost, lock-in, regulatory)

24

In multi-tenant serving for ML models at very large scale (millions of users, thousands of models), how do you design the feature-store, model-registry, serving fabric, metadata tracking, and what trade offs do you make between isolation (per-tenant models) vs model reuse vs cost?

25

You're working with a sensor-fusion system for autonomous robotics (vision + lidar + radar + ML). The deep models produce probabilistic outputs under uncertain input conditions. Describe how you design an end-to-end system (data collection, model calibration, uncertainty propagation, real-time inference, fail-safe fallback) and how you validate it for safety/performance.



C. Advanced Level

26

As model sizes continue to grow, energy & carbon cost become material. Suppose you need to justify a model choice to executive leadership. How would you build a cost-/benefit model (latency, accuracy uplift, business value, energy cost, inference cost) and how would you present trade-offs between large model vs smaller efficient model vs retrieval-augmented hybrid?

27

You're responsible for the internal tooling that tracks model lineage, drift, versioning, fairness, reproducibility across hundreds of ML projects. How would you architect the tooling, what data schema would you design, how would you enforce compliance (audit trails), and how would you surface anomalies proactively?

28

You are integrating an open-source retrieval-augmented generation (RAG) stack (vector DB + embedding model + LLM). A major customer complains about hallucinations tied to outdated docs in your corpus. Describe how you detect, triage, and remediate this (including updating retrieval index, measuring retrieval relevance, building answer-grounding metrics, fallback behaviours).

29

Imagine you have deployed an LLM in a high-stakes environment (legal advice, medical triage). A user queries out-of-domain content and you detect potential regulatory violation. How do you design guard-rails (prompt-engineering, answer filtering, human-in-loop, logging, audit) and how do you measure system performance, incidents, risk exposure over time?

30

You lead a research-to-production team building new model architectures (say modular networks, reasoning models). How do you define the productivity metrics (time to deploy new research, % of code in production, latency vs throughput, model drift), how do you iterate on research experiments, how do you move them into production safely, and how do you measure and incentivize this pipeline?



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